

Some notes on Wannier interpolation: External and internal contributions

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1 Context and notation

We consider the typical case in which WANNIER90 or a similar package has been used to generate a set of J Wannier functions (WFs) per cell. The Bloch-like functions obtained by Fourier transforms are

$$|u_s^W\rangle = \sum_{\mathbf{R}} e^{-i\mathbf{k}\cdot(\mathbf{r}-\mathbf{R}-\boldsymbol{\tau}_s)} |\mathbf{R},s\rangle \quad \text{[wft]} \quad (1)$$

I leave the $\mathbf{k}$ dependence implicit in the ket on the left and in most other $\mathbf{k}$ -space objects in these notes, except for occasional emphasis. I also label objects associated one-to-one with the WFs by $s, t, \dots$ while using labels $m, n, \dots$ to label H eigenstates or other states written as linear combinations of the Wannier states.

The introduction of the nominal position $\boldsymbol{\tau}_s$ of site s in the unit cell corresponds the “Convention I” of my book as implemented in PYTHTB. Setting all these variables to zero reproduces the conditions of Convention II, as also used in Refs. [1, 2] and [3]. If it is retained, there could still be important choices: (i) choose $\boldsymbol{\tau}_s$ at a “nominal” location, e.g., as given by the coordinates of the atom on which the WF is located; or (ii) choose $\boldsymbol{\tau}_s$ to be the corresponding Wannier center itself.

The usual procedure is to form the Hamiltonian¹

$$\mathbb{H}_{st}(\mathbf{k}) = \langle u_s^W | \hat{H}_{\mathbf{k}} | u_t^W \rangle \quad \text{[hwannier]} \quad (2)$$

where $\hat{H}_{\mathbf{k}} = e^{-i\mathbf{k}\cdot\mathbf{r}} \hat{H} e^{i\mathbf{k}\cdot\mathbf{r}}$. The double-staff notation seen in $\mathbb{H}$ is used in these notes to indicated any $\mathbf{k}$ -space $J \times J$ matrix written in the basis of the $|u_s^W\rangle$.² $\mathbb{H}_{st}(\mathbf{k})$ can be diagonalized as

$$U_{\text{tot}}^\dagger \mathbb{H} U_{\text{tot}} = H^{\text{H}} \quad \text{[diag]} \quad (3)$$

where $U_{\text{tot},sn}$ for given band index n is the J -component column vector that defines the H eigenstate in the tight-binding (TB) sense, and the band energies $E_{n\mathbf{k}}$ are just the diagonal elements of

$$H_{mn}^{\text{H}} = E_m \delta_{mn} \cdot \quad \text{[energy]} \quad (4)$$

In these notes we assume that the number of Wannier functions exceeds the number of occupied bands, and that one has identified the M occupied states at each

¹ $\mathbb{H}_{st}$ is the same as $H_{mn}^{(\text{W})}$ in Ref. [1, 2] or [3].

²This differs from Ref. [3], where essentially all matrices are written in the Wannier basis, while double-staff notation is reserved for external quantities.

k. We let the first M columns of $U_{\text{tot},sn}$ form a semi-unitary $J \times M$ rectangular matrix U_{sv} , where v labels occupied “valence” states. Similarly, we let $\bar{U}_{sc}$ be the $J \times (M - J)$ semi-unitary matrix specifying the transformation to the unoccupied states that are contained in the Wannier manifold. We adopt the practice of writing column vectors of U or $\bar{U}$, or other internal vectors of length J , in the form $\|u_n\rangle\rangle$ to emphasize that these exist only in the tight-binding context. Even a perfect knowledge of the $\|u_n(\mathbf{k})\rangle\rangle$ is inadequate for many purposes beyond simple band structures, as the embedding information encoded in the Wannier functions is not contained in these vectors. We use a corresponding doubled bra-ket notation so that, for example, the orthogonality of occupied eigenstates is written as $\langle\langle u_v \| u_{v'} \rangle\rangle = \delta_{vv'}$ (or explicitly, $\sum_s U_{sv}^* U_{sv'} = \delta_{vv'}$).

Finally, to set the context, note that some of the previous literature focuses on computing the Abelian single-band Berry connection and curvature, as for example at the Fermi surface for calculations of transport properties. Instead, these notes focus on ground-state properties of the occupied bands as a whole, and mainly in insulators where geometric properties of the occupied manifold are of interest.

1.1 Internal vs. external contributions

At this point it is possible to work just with $\mathbb{H}_{st}$, U_{sn} , E_n , etc., in a code such as PYHTTB. We refer to these quantities as constituting the “internal” information, in the sense of being internal to the TB framework. It is important to understand, however, that all other information about the Wannier functions has been lost. The missing “external” or “embedding” information is in principle available in the physical occupied Bloch states

$$|u_v^H\rangle = \sum_s |u_s^W\rangle U_{sv}, \quad \text{[phys-bloch]} \quad (5)$$

but it would be very unwieldy to have to reference these in the target TB calculation.

To recover important parts of this missing information while still working in a TB-like context, we identify some other operators whose matrix elements can also be imported along with those of H . For a general single-particle operator $\hat{O}$ (with $\hat{O}_{\mathbf{k}} = e^{-i\mathbf{k}\cdot\mathbf{r}} \hat{O} e^{i\mathbf{k}\cdot\mathbf{r}}$) that commutes with lattice translations, such as H , one finds

$$\begin{aligned} \mathbb{O}_{st}^W(\mathbf{k}) &= \langle u_s^W | \hat{O}_{\mathbf{k}} | u_t^W \rangle \\ &= \frac{1}{N} \sum_{\mathbf{R}'\mathbf{R}} e^{i\mathbf{k}\cdot(\boldsymbol{\tau}_t + (\mathbf{R} - \mathbf{R}') - \boldsymbol{\tau}_s)} \langle \mathbf{R}'s | \hat{O} | \mathbf{R}t \rangle \\ &= \sum_{\mathbf{R}} e^{i\mathbf{k}\cdot(\mathbf{R} + \boldsymbol{\tau}_t - \boldsymbol{\tau}_s)} \langle \mathbf{0}s | \hat{O} | \mathbf{R}t \rangle. \quad \text{[per-ft-a]} \end{aligned} \quad (6)$$

The factor of $1/N$ in the second line arises because the matrix element is obtained by integrating over a unit cell instead of all space. The cell-periodicity of $\hat{O}$ and the substitution of $\mathbf{R}$ for $\mathbf{R} - \mathbf{R}'$ as a summation variable gives the final line. Adopting the notation

$$\mathbf{b}_{st}(\mathbf{R}) = \mathbf{R} + \boldsymbol{\tau}_t - \boldsymbol{\tau}_s \quad \text{[bond]} \quad (7)$$

for the bond vector pointing from site $\mathbf{0}s$ to site $\mathbf{R}t$ and

$$\mathcal{O}_{st}(\mathbf{R}) = \langle \mathbf{0}s | \hat{\mathcal{O}} | \mathbf{R}t \rangle, \quad [\text{realdef}] \quad (8)$$

for the matrix element, Eq. (6) becomes just

$$\mathbb{O}_{st}^W(\mathbf{k}) = \sum_{\mathbf{R}} e^{i\mathbf{k} \cdot \mathbf{b}_{st}(\mathbf{R})} \mathcal{O}_{st}(\mathbf{R}). \quad [\text{per-ft}] \quad (9)$$

As one example, the $\mathbb{H}_{st}(\mathbf{k})$ matrix in Eq. (2) is constructed from the real-space matrix elements $H_{st}(\mathbf{R})$ in this way.

1.2 Geometric external quantities

By “geometric” quantities we mean those that depend just on the subspace spanned by the $|\mathbf{R}s\rangle$ in real space or the $|u_s^W\rangle$ in $\mathbf{k}$ space, without reference to the Hamiltonian $\hat{H}$ or other operators that depend on $\hat{H}$. Pride of place among these belongs to the position operators $\hat{x}$, $\hat{y}$, and $\hat{z}$, which however are not periodic operators, since

$$\langle \mathbf{R}'s | \hat{r}_\mu | \mathbf{R}t \rangle = \langle \mathbf{0}s | \hat{r}_\mu | (\mathbf{R} - \mathbf{R}')t \rangle + R'_\mu \delta_{\mathbf{R}'\mathbf{R}} \delta_{st}. \quad (10)$$

Nevertheless, we can define

$$X_{\mu,st}(\mathbf{R}) = \langle \mathbf{0}s | \hat{r}_\mu - \tau_{s\mu} | \mathbf{R}t \rangle, \quad [\text{r-real}] \quad (11)$$

$$W_{\mu\nu,st}(\mathbf{R}) = \langle \mathbf{0}s | (\hat{r}_\mu - \tau_{s\mu})(\hat{r}_\nu - \tau_{t\nu}) | \mathbf{R}t \rangle. \quad [\text{rr-real}] \quad (12)$$

Eq. (11) may appear to lack symmetry in the sense that the quantity subtracted from $\hat{r}_\mu$ is written as $\tau_{s\mu}$ and not $(R_\mu + \tau_{t\mu})$. However, the orthonormality of the WFs ensures that such terms are only present when $\mathbf{R} = \mathbf{0}$ and $s = t$, so these forms are equivalent. (Note that Eq. (12) is already symmetric in this sense.) Also, it is straightforward to show that

$$X_\mu(\mathbf{R}) = X_\mu^\dagger(-\mathbf{R}), \quad [\text{Xrel}] \quad (13)$$

$$W_{\mu\nu}(\mathbf{R}) = W_{\nu\mu}^\dagger(-\mathbf{R}), \quad [\text{Wrel}] \quad (14)$$

where the dagger indicates Hermitian conjugation in the Wannier indices. Note the reversal of Cartesian indices in Eq. (14).

As we shall see, the quantities defined above will allow us to obtain the external Berry connection, Berry curvature, and metric via Fourier transform formulas similar to Eq. (9).

1.3 Non-geometric external quantities

[sec: nongeom]

Depending on the application, there are other operators whose external parts should be included in the target TB calculation. For example, matrix elements of $\hat{r}_\mu \hat{H} \hat{r}_\nu$ are needed for the Wannier-interpolation calculation of orbital magnetization [3] and of the spatial dispersion terms in the optical conductivity [4]. The treatment of these operators involves some subtleties, but details are given in those references.

1.4 Special role of the position matrix elements

[sec: small]

For most operators in Eq. (8), we might expect that the diagonal elements $\langle \mathbf{0}s | \hat{O} | \mathbf{0}s \rangle$ would tend to dominate over off-diagonal elements. However, the position operators are special when working in Convention I. A diagonal element of Eq. (11) is $X_{\mu,ss}(\mathbf{0}) = \bar{r}_{s\mu} - \tau_{s\mu}$, i.e., just the separation of the Wannier center $\bar{r}_{s\mu}$ from the nominal center $\tau_{s\mu}$. Thus, when using Convention I, these diagonal elements are typically very small (when using atomic centers for the τ_s) or exactly zero (when using Wannier centers for the τ_s). Moreover, in Eq. (15) of Ref. [5] the off-diagonal elements were shown to obey

$$\tilde{\Omega}_{\text{MV}} = \sum_s \sum_{\mathbf{R}t \neq \mathbf{0}s} |\langle \mathbf{0}s | \hat{\mathbf{r}} | \mathbf{R}t \rangle|^2, \quad [\text{omega-mv}] \quad (15)$$

where $\tilde{\Omega}_{\text{MV}}$ is the non-gauge-invariant part of the Marzari-Vanderbilt spread functional [5] (not to be confused with a Berry curvature). There and subsequently it has been found that $\tilde{\Omega}_{\text{MV}}$ is typically a small fraction of the total spread, perhaps 10%, even when studying just the occupied manifold of an insulator.³

[1] A relevant empirical result was also reported in [1, 2]. There Convention II was used, but in the application to mono-atomic materials such as Fe and Ni studied there, with the atom on the origin, the two conventions are indistinguishable. [Wang 2006] were calculating the anomalous Hall conductivity of Fe, and found that the external terms, which represent the presence of nonzero $\langle \mathbf{0}s | \hat{\mathbf{r}} | \mathbf{R}t \rangle$ terms, accounts for only about 1% of the results.

DV[1]: Did we report the size of the external terms for Co, where there are two atoms per cell and Convention II ought to be worse?

We take it as an interesting working hypothesis that the external Berry connection, and other geometric quantities derivable from it, such as the Berry curvature, provide only a small correction to the internal terms. This hypothesis seems worthy of testing in a variety of materials settings.

We emphasize that the same argument does not apply to the quantum metric, which relates to $W_{\mu\nu,st}(\mathbf{R})$, or to the operators considered in Sec. 1.3, where there is no obvious reason for the diagonal elements to be small.

1.5 External geometric quantities in $\mathbf{k}$ space

[sec: exgeom]

The basic building blocks of geometric quantities in $\mathbf{k}$ space are

$$\mathbb{A}_{\mu,st}(\mathbf{k}) = \langle u_s^{\text{W}} | i\partial_{\mu} u_t^{\text{W}} \rangle \quad [\text{AA}], \quad (16)$$

$$\mathbb{W}_{\mu\nu,st}(\mathbf{k}) = \langle \partial_{\mu} u_s^{\text{W}} | \partial_{\nu} u_t^{\text{W}} \rangle \quad [\text{FF}] \quad (17)$$

corresponding to the Berry connection and metric-curvature tensors respectively. Using Eq. (1) and the definitions in Eqs. (11) and (12), it is straightforward to show

³We have in mind a typical case in which one chooses complete shells of p or d states for the Wannier content, which is closer to an atomic limit.

that

$$\mathbb{A}_{\mu,st}(\mathbf{k}) = \sum_{\mathbf{R}} e^{i\mathbf{k}\cdot\mathbf{b}_{st}(\mathbf{R})} X_{\mu,st}(\mathbf{R}), \quad [\text{ft-aa}] \quad (18)$$

$$\partial_{\mu}\mathbb{A}_{\nu,st}(\mathbf{k}) = i \sum_{\mathbf{R}} e^{i\mathbf{k}\cdot\mathbf{b}_{st}(\mathbf{R})} b_{\mu,st}(\mathbf{R}) X_{\nu,st}(\mathbf{R}), \quad [\text{ft-daa}] \quad (19)$$

$$\mathbb{W}_{\mu\nu,st}(\mathbf{k}) = \sum_{\mathbf{R}} e^{i\mathbf{k}\cdot\mathbf{b}_{st}(\mathbf{R})} W_{\mu\nu,st}(\mathbf{R}). \quad [\text{ft-ff}] \quad (20)$$

For convenience we also define the curl of the connection

$$\Omega_{\mu\nu,st}(\mathbf{k}) = \partial_{\mu}\mathbb{A}_{\nu,st}(\mathbf{k}) - \partial_{\nu}\mathbb{A}_{\mu,st}(\mathbf{k}), \quad [\text{curlk}] \quad (21)$$

which is essentially the external non-covariant Berry curvature of the Wannier manifold, and is easily evaluated using Eq. (19). As our focus in these notes will be on geometric quantities, these are the basic ingredients that we will be making use of when computing the external contributions to quantities of interest. As a reminder, Eqs. (18) and (19) are subject to the smallness hypothesis of Sec. 1.4, while Eq. (20) is not.

From Eqs. (13-14) it follows that $\mathbb{A}_{\mu}(\mathbf{k}) = \mathbb{A}_{\mu}^{\dagger}(\mathbf{k})$ and $\mathbb{W}_{\mu\nu}(\mathbf{k}) = \mathbb{W}_{\nu\mu}^{\dagger}(\mathbf{k})$. Thus, $\mathbb{A}_{\mu}$ is Hermitian in the Wannier indices for given Cartesian indices, but $\mathbb{W}_{\mu\nu}$ is not. As it happens, the antihermitian part of $\mathbb{W}_{\mu\nu}$ is actually redundant with $\Omega_{\mu\nu}$:

$$\Omega_{\mu\nu} = i\mathbb{W}_{\mu\nu} - i\mathbb{W}_{\nu\mu}. \quad [\text{CtoW}] \quad (22)$$

This follows from $\partial_{\mu}\mathbb{A}_{\nu,st} = \partial_{\mu}\langle u_s^W | i\partial_{\nu}u_t^W \rangle = i\langle \partial_{\mu}u_s^W | \partial_{\nu}u_t^W \rangle + i\langle u_s^W | \partial_{\mu}\partial_{\nu}u_t^W \rangle$, and similarly for $\partial_{\nu}\mathbb{A}_{\mu}$, leading to a cancellation of the second terms when plugged into Eq. (21). Similarly, we denote the Hermitian or “quadrupole” part of $\mathbb{W}_{\mu\nu}$ by

$$\mathbb{Q}_{\mu\nu} = \frac{\mathbb{W}_{\mu\nu} + \mathbb{W}_{\nu\mu}}{2}, \quad [\text{quad}] \quad (23)$$

where “quadrupole” is understood here to include the trace part.⁴

2 Geometric quantities in $\mathbf{k}$ space

Consider a portion of the BZ in which there are M occupied bands ($M < J$) described by a smooth gauge of states $|u_v\rangle$ that are generally not eigenstates of H . That is, the $|u_v\rangle$ span the same M -dimensional subspace as the $|u_v^H\rangle$. In a metal this could be a portion of the BZ not intersected by a Fermi surface, or in an insulator it could be the entire BZ. We assume that the $|u_v\rangle$ have been constructed as

$$|u_v\rangle = \sum_s^J |u_s^W\rangle V_{sv}, \quad [\text{vdef}] \quad (24)$$

⁴Note that the Brillouin zone integral of the Cartesian and band trace of $\mathbb{Q}(\mathbf{k})$ is, up to a constant, just $\sum_{\mu s} W_{\mu\mu,ss}(\mathbf{0})$ in the notation of Eq. (12). This corresponds to the total spread of the Wannier functions (all J of them) as defined in Eq. (11) of Ref. [5].

where V is a $J \times M$ semi-unitary matrix, just like U in Eq. (5). In these notes, we will be working mainly in the framework of this putative smooth gauge.

To complete the relations between the three basis sets, we define the $M \times M$ unitary matrix Y via

$$|u_v^H\rangle = \sum_{v'}^M |u_{v'}\rangle Y_{v'v}, \quad [\text{ydef}] \quad (25)$$

so that $U = VY$. Note that

$$U^\dagger U = V^\dagger V = Y^\dagger Y = Y Y^\dagger = 1_{M \times M} \quad [\text{ones}] \quad (26)$$

(the $M \times M$ identity matrix), but that $UU^\dagger = VV^\dagger$ are *not* equal to the $J \times J$ identity. In fact, it is useful to define for later use a $J \times J$ matrix

$$\mathbb{P}_{st} = \langle u_s^W | \hat{P} | u_t^W \rangle = (UU^\dagger)_{st} = (VV^\dagger)_{st}, \quad [\text{fdef}] \quad (27)$$

where the band projector onto the manifold of occupied states can be written in terms of either set of states as

$$\hat{P} = \sum_v^M |u_v^H\rangle \langle u_v^H| = \sum_v^M |u_v\rangle \langle u_v|. \quad [\text{bproj}] \quad (28)$$

Note that $\mathbb{P}$, which is the same as the f matrix introduced by [Lopez], has the interpretation of a density matrix in the Wannier basis. In any case, it is important to remember that U and V are only semi-unitary, and manipulations commonly used for unitary matrices do not necessarily apply to them.

To summarize, we assume the existence of an “explicit smooth gauge” of states $|u_v\rangle$ spanning the occupied bands space, as specified by V_{sv} . We shall assume moreover that the derivatives $\partial_\mu V_{sv}$ are also known. A procedure for constructing such an explicit smooth gauge, and for computing $\partial_\mu V$, has recently been introduced by [Cole]. As we shall see, we will not need this explicit smooth gauge for constructing many gauge-invariant quantities of interest, as for example the band projector in Eq. (28). However, it may be needed when it comes to computing a Berry phase or axion angle as a BZ integral of a gauge-dependent quantity.

2.1 Berry connection in 1D

[sec: conn-1d]

Working in the smooth gauge, the $M \times M$ Berry connection matrix takes the form

$$A_{\mu, vv'}(\mathbf{k}) = \langle u_v | i \partial_\mu u_{v'} \rangle. \quad [\text{b-conn}] \quad (29)$$

From Eq. (24) we find

$$|\partial_\mu u_v\rangle = \sum_s^J |u_s^W\rangle (\partial_\mu V_{sv}) + \sum_s^J |\partial_\mu u_s^W\rangle V_{sv}. \quad [\text{dketu}] \quad (30)$$

We identify the terms on the right as “internal” and “external” contributions, respectively. Plugging Eq. (24) in again for the bra in Eq. (29) and using Eq. (16), we get

$$A_{\mu, vv'} = A_{\mu, vv'}^{\text{I}} + A_{\mu, vv'}^{\text{E}}, \quad [\text{Asum}] \quad (31)$$

where matrices

$$A_{\mu}^{\text{I}} = V^{\dagger} i \partial_{\mu} V, \quad [\text{AIdef}] \quad (32)$$

$$A_{\mu}^{\text{E}} = V^{\dagger} \mathbb{A}_{\mu} V \quad [\text{AEdef}] \quad (33)$$

denote the internal and external terms respectively. The Berry connection of Eq. (31) will be a basic building block in what follows in later sections.

For the purpose of computing electric polarization in an insulator, one needs the traced Berry connection appearing in the integrand of

$$\phi_{\mu} = \oint dk_{\mu} \text{tr} [A_{\mu}^{\text{I}} + A_{\mu}^{\text{E}}], \quad [\text{bph}] \quad (34)$$

where the lower-case ‘tr’ refers to a trace over occupied space of dimension M . We work in the context of a one-dimensional crystal in the remainder of this subsection. Equation (34) is the desired expression in the occupied-space representation.

Alternatively, we can work in the Wannier representation. Note that, for any $J \times J$ matrix $\mathbb{O}$, $\text{tr} [V^{\dagger} \mathbb{O} V] = \text{Tr} [V V^{\dagger} \mathbb{O}] = \text{Tr} [\mathbb{p} \mathbb{O}]$, where $\mathbb{p}$ is the density matrix defined in Eq. (27), and the upper-case ‘Tr’ indicates a trace over the Wannier space of dimension J . Thus, $\text{tr} A_{\mu}^{\text{E}} = \text{Tr} [\mathbb{p} \mathbb{A}_{\mu}]$, etc. Using this, we find

$$\phi_{\mu} = \oint dk_{\mu} \text{Tr} [\mathbb{p} (\mathfrak{a}_{\mu} + \mathbb{A}_{\mu})] \quad [\text{bphase}] \quad (35)$$

where

$$\mathfrak{a}_{\mu} = i(\partial_{\mu} V) V^{\dagger}. \quad [\text{bbma}] \quad (36)$$

(Note that while $A_{\mu}^{\text{I}} = V^{\dagger} \mathfrak{a}_{\mu} V$, instead $\mathfrak{a}_{\mu} \neq V A_{\mu}^{\text{I}} V^{\dagger}$, and while A_{μ}^{I} is Hermitian, $\mathfrak{a}_{\mu}$ is not. These observations follow from the semiunitary nature of V .) The two terms in Eq. (35) are the internal and external terms respectively in a Wannier-space formulation like the one introduced by [3].

The Berry-phase expressions in Eqs. (34) and (35) suffer from the difficulty that the first term in the integrand is not gauge invariant. While the occupied subspace projector $\mathbb{p}$ introduced in Eq. (27) is gauge-covariant, $\mathfrak{a}_{\mu}$ is not. As a result, ϕ_{μ} is ill-defined in a metal. However, in an insulator, the first term becomes gauge-invariant modulo 2π after $\text{Tr} [\mathbb{p} \mathfrak{a}_{\mu}]$ is integrated over the full BZ, as will be discussed shortly. The external term $\phi_{\mu}^{\text{E}} = \oint dk_{\mu} \text{Tr} [\mathbb{p} \mathbb{A}_{\mu}]$ in Eq. (35) is immune to these issues; its gauge-invariance follows from the gauge-covariance of $\mathbb{A}_{\mu}$. Thus, one can avoid the need to reference the smooth gauge, and it could equally well be evaluated in the Hamiltonian gauge. Essentially, ϕ_{μ}^{E} is just an “expectation of $\hat{x}$ ” computed by

tracing the X_μ operator of Eq. (11) against the density matrix, and is expected to be a small correction when using Convention I.

The evaluation of the internal (first) term in Eq. (35), on other hand, does appear to require an explicit choice of smooth and periodic gauge as encoded in the V_{sn} . In fact, if such an explicit gauge is available, and if its Berry connection in Eq. (32) is known [6], the internal term is also easily computed. The Berry phase may then be evaluated as a simple sum over a mesh of k -points along the 1D BZ. Alternatively, one can adopt any gauge, as for example the Hamiltonian gauge, and evaluate ϕ_μ^I using the “modern theory of polarization” [7]. That is, defining the inner product matrices $M_{vv'}^{(k,k+\delta k)} = \langle\langle u_{vk}^H | u_{v',k+\delta k}^H \rangle\rangle$, where $|u_{vk}^H\rangle$ is a column vector of U_k , the Berry phase is given by the standard expression

$$\phi_\mu^I = -\text{Im} \ln \det \prod_k M^{(k,k+\delta k)} \quad \text{[discrete-berry]} \quad (37)$$

of the modern theory approach.

So, for the case of electric polarization, both the dominant internal term and the small external correction can be obtained in a straightforward manner, without the need for an explicit smooth gauge. However, such a smooth gauge is still needed if the full non-Abelian Berry connection matrix $A_{\mu,vv'}$ of Eq. (31) is required for constructing other objects. We shall encounter a situation like this in Sec. 2.3.

2.2 Berry curvature and metric in 2D

[sec: bc-2d]

General formulation. The information about the Berry curvature and metric tensors is contained in the gauge-covariant metric-curvature tensor. Working again in the smooth gauge, this is defined as

$$F_{\mu\nu,vv'} = \langle \partial_\mu u_v | \hat{Q} | \partial_\nu u_{v'} \rangle \quad \text{[Fa]} \quad (38)$$

where $\hat{Q} = 1 - \hat{P} = 1 - \sum_v^M |u_v\rangle\langle u_v|$. To evaluate $F_{\mu\nu}$, we adopt the notation $\hat{Q} = \hat{Q}^I + \hat{Q}^E$, where $\hat{Q}^I$ projects onto unoccupied states in the Wannier manifold, and $\hat{Q}^E$ projects onto all states outside the Wannier manifold. Then

$$F_{\mu\nu,vv'} = \langle \partial_\mu u_v | \hat{Q}^I | \partial_\nu u_{v'} \rangle + \langle \partial_\mu u_v | \hat{Q}^E | \partial_\nu u_{v'} \rangle. \quad \text{[Fb]} \quad (39)$$

We also rewrite Eq. (30) as

$$|\partial_\mu u_v\rangle = -i \sum_s^J |u_s^W\rangle (\mathfrak{a}_\mu V)_{sv} + \sum_s^J |\partial_\mu u_s^W\rangle V_{sv}, \quad \text{[dku]} \quad (40)$$

where we have used that $\partial_\mu V = -i\mathfrak{a}_\mu V$ from Eq. (36). Plugging Eq. (40) into (39), we find that

$$F_{\mu\nu} = F_{\mu\nu}^I + F_{\mu\nu}^X + F_{\mu\nu}^E \quad \text{[Fsum]} \quad (41)$$

where, defining $\mathfrak{q} = \mathbb{1} - \mathfrak{p}$,

$$F_{\mu\nu}^{\text{I}} = V^\dagger \mathfrak{a}_\mu^\dagger \mathfrak{q} \mathfrak{a}_\nu V, \quad [\text{FI}] \quad (42)$$

$$F_{\mu\nu}^{\text{X}} = V^\dagger \mathfrak{a}_\mu^\dagger \mathfrak{q} \mathbb{A}_\nu V + V^\dagger \mathbb{A}_\mu \mathfrak{q} \mathfrak{a}_\nu V, \quad [\text{FX}] \quad (43)$$

$$F_{\mu\nu}^{\text{E}} = V^\dagger (\mathbb{A}_\mu \mathfrak{q} \mathbb{A}_\nu + \mathbb{W}_{\mu\nu} - \mathbb{A}_\mu \mathbb{A}_\nu) V \quad [\text{FE}]. \quad (44)$$

Inserting Eq. (40) into the first term of Eq. (39) gives rise to Eqs. (42-43) (internal and cross terms respectively), and to the first term of Eq. (44) (external terms). In the second term of Eq. (39), however, only the second term of Eq. (40) enters, since $\hat{Q}^{\text{E}}$ annihilates $\langle u_s^{\text{W}} |$ or $| u_t^{\text{W}} \rangle$. As a result, the second term of Eq. (39) gives rise to the last two terms of Eq. (44). To see this, note that

$$\begin{aligned} \langle \partial_\mu u_s^{\text{W}} | \hat{Q}^{\text{E}} | \partial_\nu u_t^{\text{W}} \rangle &= \langle \partial_\mu u_s^{\text{W}} | \partial_\nu u_t^{\text{W}} \rangle - \sum_r^J \langle \partial_\mu u_s^{\text{W}} | u_r^{\text{W}} \rangle \langle u_r^{\text{W}} | \partial_\nu u_t^{\text{W}} \rangle \\ &= (\mathbb{W}_{\mu\nu} - \mathbb{A}_\mu \mathbb{A}_\nu)_{st}. \quad [\text{FEintermed}] \end{aligned} \quad (45)$$

Thus we obtain that $F_{\mu\nu} = V^\dagger \mathbb{F}_{\mu\nu} V$, where⁵ [2]

$$\mathbb{F}_{\mu\nu} = (\mathfrak{a}_\mu^\dagger + \mathbb{A}_\mu) \mathfrak{q} (\mathfrak{a}_\nu + \mathbb{A}_\nu) + \mathbb{W}_{\mu\nu} - \mathbb{A}_\mu \mathbb{A}_\nu. \quad [\text{FEfin}] \quad (46)$$

We have access to $\mathbb{A}_\mu$ and $\mathbb{W}_{\mu\nu}$ from Sec. 1.5, and $\mathfrak{p}$ and $\mathfrak{q}$ are easily constructed. As for $\mathfrak{a}_\mu$, recall that we have assumed that V and its derivatives $\partial_\mu V$ are known, which implies that A_μ^{I} of Eq. (32) is also known. However, $\mathfrak{a}_\mu$ and A_μ^{I} are related by $A_\mu^{\text{I}} = V^\dagger \mathfrak{a}_\mu V$, which cannot be inverted to obtain $\mathfrak{a}_\mu$. In fact, the information needed to complete the knowledge of $\mathfrak{a}_\mu$ concerns precisely the cross matrix elements of the internal connection between occupied and (Wannier-space) unoccupied states, as needed, for example, when evaluating terms like $\mathfrak{q} \mathfrak{a}_\mu V$. However, these matrix elements are easily obtained by perturbation theory in the Hamiltonian gauge. Consider, for example, the $\mathbb{A}_\mu \mathfrak{q} \mathfrak{a}_\nu$ term in Eq. (46), which gives rise to a term in $F_{\mu\nu, vv'}$ that can be written as

$$(V^\dagger \mathbb{A}_\mu \mathfrak{q} \mathfrak{a}_\nu V)_{vv'} = \sum_c^L (V^\dagger \mathbb{A}_\mu \bar{U})_{vc} (\bar{U}^\dagger i \partial_\nu V)_{cv'} = (\bar{A}_\mu^{\text{E}\dagger} \bar{A}_\nu^{\text{I}})_{vv'}. \quad [\text{aAterm}] \quad (47)$$

Here $L = J - M$ has been introduced for the number of unoccupied states in the Wannier manifold. As a reminder, in our notation index c always refers to one of the L unoccupied eigenstates inside the Wannier space, while v and v' always belong to one of the M states (not necessarily eigenstates) in the occupied manifold. The barred quantities in Eq. (47) are

$$\bar{U}_{sc} = \langle u_s^{\text{W}} | u_c^{\text{H}} \rangle, \quad [\text{Ubar}] \quad (48)$$

$$\bar{A}_{\mu, cv}^{\text{I}} = (\bar{U}^\dagger i \partial_\mu V)_{cv}, \quad [\text{AIbar}] \quad (49)$$

$$\bar{A}_{\mu, cv}^{\text{E}} = (\bar{U}^\dagger \mathbb{A}_\mu V)_{cv}, \quad [\text{AEbar}] \quad (50)$$

⁵Note that the trace of this quantity is consistent with Eq. (48) of Ref. [3], where the order of terms is reversed and our $\mathbb{W} - \mathbb{A}\mathbb{A}$ is their $\tilde{\mathbb{F}}$.

DV[2]: I added a handwritten note in the Overleaf file tree that makes a direct comparison with the tensor, not just with its trace. See 'lopez-comparison.pdf'.

defined in analogy to similar quantities introduced earlier, but now connecting to unoccupied states $|u_c^H\rangle$ in $\hat{Q}^I$ (i.e., in the support of $\mathfrak{q}$) instead of occupied states. These are rectangular matrices; $\bar{U}$ is $J \times L$ while $\bar{A}^I$ and $\bar{A}^E$ are $L \times M$.

Starting from Eq. (49) and making use of $V = UY^\dagger$, it follows that $\bar{U}^\dagger \partial_\mu V = \bar{U}^\dagger (\partial_\mu U) Y^\dagger + \bar{U}^\dagger U \partial_\mu Y^\dagger$, but the second term vanishes since $(\bar{U}^\dagger U)_{cv} = \langle\langle u_c^H \| u_v^H \rangle\rangle = 0$. It follows that

$$\bar{A}_\mu^I = \bar{A}_\mu^{I,H} Y^\dagger, \quad [\text{AbarI}] \quad (51)$$

where

$$\bar{A}_{\mu,cv}^{I,H} = \bar{U}^\dagger (\partial_\mu U) = \langle\langle u_c^H \| i \partial_\mu u_v^H \rangle\rangle. \quad [\text{Adetails}] \quad (52)$$

Equation (51) reflects the gauge-covariance of $\bar{A}_\mu^I$, as expected for an interband (conduction-valence) Berry connection. Using ordinary perturbation theory,

$$\bar{A}_{\mu,cv}^{I,H} = i \frac{\langle\langle u_c^H \| \partial_\mu \mathbb{H} \| u_v^H \rangle\rangle}{E_v - E_c}, \quad [\text{veloc}] \quad (53)$$

where the numerator on the right is essentially a cross-gap velocity matrix element. This gives us access to the cross-gap connection in our working gauge. We can now sum the internal and external cross-gap connections of Eqs. (51) and (50) to obtain the total

$$\bar{A}_\mu = \bar{A}_\mu^I + \bar{A}_\mu^E. \quad [\text{Abar}] \quad (54)$$

Returning now to Eq. (46), it follows that the contribution to $F_{\mu\nu}$ from the first term is just $\bar{A}_\mu^\dagger \bar{A}_\nu$. Taking note of Eq. (33) and making the parallel definition

$$W_{\mu\nu}^E = V^\dagger \mathbb{W}_{\mu\nu} V, \quad [\text{WEdef}] \quad (55)$$

the contribution to $F_{\mu\nu}$ from the second term of Eq. (46) is just $W_{\mu\nu}^E$. Finally, the last term in Eq. (46) gives rise to a contribution

$$\begin{aligned} -V^\dagger \mathbb{A}_\mu \mathbb{A}_\nu V &= -V^\dagger \mathbb{A}_\mu (\mathbb{P} + \mathfrak{q}) \mathbb{A}_\nu V \\ &= -(V^\dagger \mathbb{A}_\mu V) (V^\dagger \mathbb{A}_\nu V - (V^\dagger \mathbb{A}_\mu \bar{U})(\bar{U}^\dagger \mathbb{A}_\nu V)) \\ &= -A_\mu^E A_\nu^E - (\bar{A}_\mu^E)^\dagger \bar{A}_\nu^E \quad [\text{newAA}] \end{aligned} \quad (56)$$

where in the second line we have used that $VV^\dagger = \mathbb{P}$ and $\bar{U}\bar{U}^\dagger = \mathfrak{q}$. Assembling the results, we finally arrive at

$$F_{\mu\nu} = \bar{A}_\mu^\dagger \bar{A}_\nu - (\bar{A}_\mu^E)^\dagger \bar{A}_\nu^E + W_{\mu\nu}^E - A_\mu^E A_\nu^E. \quad [\text{Fgood}] \quad (57)$$

In view of Eq. (54), note that the first two terms above include the II, IE, and EI terms, but not the purely external EE terms.

Equation (57) is one of the major results of these notes. [\[3\]](#) Note that the valence-valence Berry connection encoded in A_μ^I has disappeared from the expression in

DV[\[3\]](#): Agrees with previous expressions in the literature?
IS: Need to look back at the preprint with Stepan...

Eq. (57). This means that, for the purposes of computing the Berry curvature or metric, and also for the four-curvature in Sec. 2.4, there is no advantage to working in the smooth gauge. One can evaluate these expressions directly in the Hamiltonian gauge; since only cross-gap and external matrix elements appear, any nonanalytic behavior of the Hamiltonian-gauge wave functions will not cause problems.

In the absence of any external terms, Eq. (57) becomes just $F_{\mu\nu} = Y F_{\mu\nu}^{\text{I,H}} Y^\dagger$ with

$$F_{\mu\nu, vv'}^{\text{I,H}} = \sum_c^L \frac{\langle\langle u_v^{\text{H}} | (\partial_\mu \mathbb{H}) | u_c^{\text{H}} \rangle\rangle \langle\langle u_c^{\text{H}} | (\partial_\nu \mathbb{H}) | u_{v'}^{\text{H}} \rangle\rangle}{(E_v - E_c)(E_{v'} - E_c)}. \quad [\text{kubo}] \quad (58)$$

This is the standard Kubo expression.

Berry curvature. In terms of F , the gauge-covariant Berry curvature tensor is

$$B_{\mu\nu, vv'} = iF_{\mu\nu, vv'} - iF_{\mu\nu, v'v}^*. \quad [\text{Fcurv}] \quad (59)$$

Since F is Hermitian in the joint indices, $F_{\mu\nu, vv'} = F_{\nu\mu, v'v}^*$, this can also be written

$$B_{\mu\nu} = iF_{\mu\nu} - iF_{\nu\mu}. \quad [\text{Fscurv}] \quad (60)$$

When plugging Eq. (57) into this expression, it is evident that only the antihermitian part of $W_{\mu\nu}^{\text{E}}$ enters, and according to Eq. (22) this can be replaced by the curl of connection, so that

$$B_{\mu\nu} = i[\bar{A}_\mu^\dagger \bar{A}_\nu - (\bar{A}_\mu^{\text{E}})^\dagger \bar{A}_\nu^{\text{E}}] + \text{h.c.} + \Omega_{\mu\nu}^{\text{E}} - i[A_\mu^{\text{E}}, A_\nu^{\text{E}}]. \quad [\text{Bgood}] \quad (61)$$

where we have defined

$$\Omega_{\mu\nu}^{\text{E}} = V^\dagger \Omega_{\mu\nu} V \quad [\text{SEdef}] \quad (62)$$

in terms of Ω from Eq. (21), in analogy with Eq. (55).

The first two terms in Eq. (61) for the gauge-covariant Berry curvature take a Kubo form in which the cross-gap Berry connections $\bar{A}_{\mu, cv}$ contain purely internal terms and external-internal cross terms, but no purely external terms. The last two terms are purely external and correspond to the standard expression for the gauge-covariant Berry curvature of the full Wannier space, but projected onto the occupied subspace. Since the contributions from Eq. (20) have disappeared, only the X_μ of Eq. (11), and not the $W_{\mu\nu}$ of Eq. (12), are needed for calculating the Berry curvature.

For the purposes of computing the intrinsic anomalous Hall conductivity, we only need the band trace of the Berry curvature. Under the trace, the commutator in Eq. (61) vanishes, so that⁶

$$\text{tr}[B_{\mu\nu}] = -2 \text{Im tr}[\bar{A}_\mu^\dagger \bar{A}_\nu - (\bar{A}_\mu^{\text{E}})^\dagger \bar{A}_\nu^{\text{E}}] + \text{tr}[\Omega_{\mu\nu}^{\text{E}}]. \quad [\text{trBcurv}] \quad (63)$$

⁶After converting notation, this is readily seen to be consistent with Eq. (51) of Ref. [3].

In principle, it should be possible to use these formulas to compute quantities for which indices like μ live in (k, β) space, where β is a parameter of the Hamiltonian. For example, the Born effective charge in direction μ can be expressed in terms of a Berry curvature in (k_μ, β) space, where β is an atomic displacement. The same applies to the computation of the dynamical axion response to a phonon perturbation in (k_x, k_y, k_z, β) space as described in Sec. 2.4. However, some of the formulas need to be modified when constructing quantities involving a parametric derivative. These issues are explored in the Appendix.

Metric tensor. In terms of F , the gauge-covariant metric tensor is

$$R_{\mu\nu} = \frac{1}{2}(F_{\mu\nu} + F_{\nu\mu}), \quad [\text{Fsmet}] \quad (64)$$

and upon inserting Eq. (57) we find

$$R_{\mu\nu} = \frac{1}{2}[\bar{A}_\mu^\dagger \bar{A}_\nu - (\bar{A}_\mu^\dagger)^\dagger \bar{A}_\nu^\dagger] + \text{h.c.} + Q_{\mu\nu}^E - \frac{1}{2}\{A_\mu^E, A_\nu^E\} \quad [\text{Rmunu}] \quad (65)$$

where

$$Q_{\mu\nu}^E = V^\dagger Q_{\mu\nu} V \quad [\text{QEdef}] \quad (66)$$

is the projected external quadrupole contribution. The traced metric simplifies slightly to become

$$\text{tr}[R_{\mu\nu}] = \text{Re tr}[\bar{A}_\mu^\dagger \bar{A}_\nu - (\bar{A}_\mu^\dagger)^\dagger \bar{A}_\nu^\dagger + Q_{\mu\nu}^E - A_\mu^E A_\nu^E]. \quad [\text{trMetric}] \quad (67)$$

This time it is essential to import the second-moment matrix elements of Eq. (12), since they enter $Q_{\mu\nu}^E$. This provides a reminder that, unlike for other geometric quantities considered in these notes, the metric is not dominated by internal terms, even in TB Convention I. As an extreme case, a lattice of isolated He atoms will have negligible internal contribution to the metric, which will be dominated by the diagonal matrix elements $Q_{\mu\mu}$ which measure the size of the atomic He s orbitals.

2.3 Chern-Simons 3-form and axion coupling in 3D

[sec: axion]

As was the case for the Berry phase in 1D, the axion angle in 3D requires a smooth and periodic gauge, and this time there is no known formulation analogous to the Wilson loop that allows us to avoid the actual construction of a smooth gauge. We assume this has been done using the methods of Ref. [6], and we restrict ourselves to insulators here.

The axion angle is given by

$$\theta = -\frac{1}{4\pi} \int_{\text{BZ}} d^3k \epsilon_{\mu\nu\sigma} \text{tr} \left[\frac{1}{2} A_\mu B_{\nu\sigma} + \frac{i}{3} A_\mu A_\nu A_\sigma \right] \quad [\text{axion}] \quad (68)$$

where $B_{\mu\nu}$ is again the full gauge-covariant Berry curvature tensor of Eq. (61), and A_μ is the full Berry connection of Eq. (31) containing both internal and external terms. The procedure for computing the axion angle is thus basically that of Ref. [6], except that the external terms are included before assembling the result according to Eq. (68).

2.4 Four-curvature in 4D

[sec: four-curv]

The change of θ under adiabatic evolution is given by

$$\theta(\beta) - \theta(0) = \frac{1}{16} \int_0^\beta d\beta' \int_{\text{BZ}} d^3k \epsilon_{\mu\nu\sigma\tau} \text{tr} [B_{\mu\nu} B_{\sigma\tau}] \quad [\text{del-theta}] \quad (69)$$

where we are working in $(\mu, \nu, \sigma, \tau) = (k_x, k_y, k_x, \beta)$ space. Once again, the ingredients needed to evaluate Eq. (69) are available from Sec. 2.2. However, the WFs $|u_s^{\text{W}}\rangle$ now vary with β , leading to some modifications to the formulation as detailed in the Appendix.

2.5 Discussion

These notes have focused on the context in which the Wannier construction is done so as to span a space including some unoccupied bands, typically using the disentangling approach of Ref. [8]. In the opposite case that the occupied space of an insulator is directly Wannierized, so that the Wannier and occupied spaces coincide, all internal quantities vanish identically and only the purely external terms contributions survive. A calculation of this kind was done for the axion coupling of Cr_2O_3 and Bi_2Se_3 , but working entirely in real space [9].

3 Parametric derivatives

We now assume that the Hamiltonian has an additional dependence on some parameter β which we take here to be an atomic displacement, although it could be any perturbation. We may wish to use Wannier interpolaton to compute the dynamical effective charge in direction μ , which requires a BZ integral of the mixed Berry curvature $\Omega_{\beta\mu}$, or a derivative of the axion coupling, which involves similar quantities. We can extend the results obtained above to a formulation in which indices such as μ run over (k_x, k_y, k_z, β) , with all four indices treated in a symmetric fashion. However, some of the formulas appearing above have to be modified when constructing quantities carrying a β index.

Specifically, concerning external quantities, the Berry curvature analogous to Eqs. (16) and (18) becomes, after a few lines of algebra,

$$\mathbb{A}_{\beta,st}(\mathbf{k}) = \langle u_s^{\text{W}} | i\partial_\beta u_t^{\text{W}} \rangle = i \sum_{\mathbf{R}} e^{i\mathbf{k} \cdot \mathbf{b}_{st}(\mathbf{R})} \Lambda_{st}(\mathbf{R}) \quad [\text{AAb}] \quad (70)$$

where

$$\Lambda_{st}(\mathbf{R}) = \langle 0s | \partial_\beta | \mathbf{R}t \rangle. \quad [\text{Lambda}] \quad (71)$$

In deriving this equation, we require that the nominal position τ_s associated with

$|\mathbf{0}s\rangle$ be kept independent of β .⁷ Similarly, Eq. (21) becomes

$$\begin{aligned}\Omega_{\beta\mu,st}(\mathbf{k}) &= \partial_\beta \mathbb{A}_{\mu,st} - \partial_\mu \mathbb{A}_{\beta,st} \\ &= \sum_{\mathbf{R}} e^{i\mathbf{k}\cdot\mathbf{b}_{st}(\mathbf{R})} [\partial_\beta X_{\mu,st}(\mathbf{R}) + b_{\mu,st}(\mathbf{R}) \Lambda_{st}(\mathbf{R})] \cdot [\text{curl}\mathbf{k}\mathbf{b}]\end{aligned}\quad (72)$$

It is straightforward to check that the matrices in Eqs. (70) and (72) are Hermitian.

We also need the internal contribution to the Berry connection encoded in $\bar{A}_{\beta,cv}^I$ analogous to Eq. (49). This is given by Eq. (53) with the substitution $\mu \rightarrow \beta$ and $\partial_\beta \mathbb{H}(k)$ tracing back through Eq. (8) to the parametric derivative $\partial_\beta \langle \mathbf{0}s | H | \mathbf{R}t \rangle$ of the real-space Wannier Hamiltonian.

With these quantities in hand, it should now be possible to treat the β index just like the other Cartesian indices when inserted into equations such as Eqs. (33), (35), and (50). To summarize, the additional quantities needed when parametric derivatives appear are

- $\partial_\beta H_{st}(\mathbf{R}) = \partial_\beta \langle \mathbf{0}s | H | \mathbf{R}t \rangle$
- $\partial_\beta X_{\mu,st}(\mathbf{R}) = \partial_\beta \langle \mathbf{0}s | \hat{r}_\mu | \mathbf{R}t \rangle$
- $\Lambda_{st}(\mathbf{R}) = \langle \mathbf{0}s | \partial_\beta | \mathbf{R}t \rangle$

The first two can easily be obtained by finite differences between the imported data for the undistorted and distorted structures. The last is trickier; after Fourier transformation back to $\mathbf{k}$ space, it involves inner products of Bloch states at two different structures.

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⁷Recall that we do not require that the τ_s coincide with the Wannier centers; they can be chosen freely as long as the choice is used consistently. Here we require only that these nominal centers are kept identical for undistorted and distorted structures. This assumption can be relaxed at the expense of requiring additional terms in some formulas.

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